

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF
ENGINEERING)**

Department of Artificial Intelligence and Data Science

DSA0503 – Query Processing For Data Science

CO1_AT2_Application based questions

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1. Electricity Bill Generation

An electricity department maintains consumer information in JSON format and monthly meter readings in CSV format. Write a Python program to calculate electricity bills based on the readings and generate a billing report for all consumers.

Aim

To calculate electricity bills using consumer details from a JSON file and monthly meter readings from a CSV file.

Procedure

1. Read consumer details from the JSON file.
2. Read monthly electricity units from the CSV file.
3. Calculate the electricity bill based on the number of units consumed.
4. Display the billing report for all consumers.

Input

consumers.json

```
[  
  {"id":101,"name":"Arun","address":"Chennai"},  
  {"id":102,"name":"Priya","address":"Coimbatore"},  
  {"id":103,"name":"Rahul","address":"Madurai"}  
]
```

readings.csv

id,units

101,120

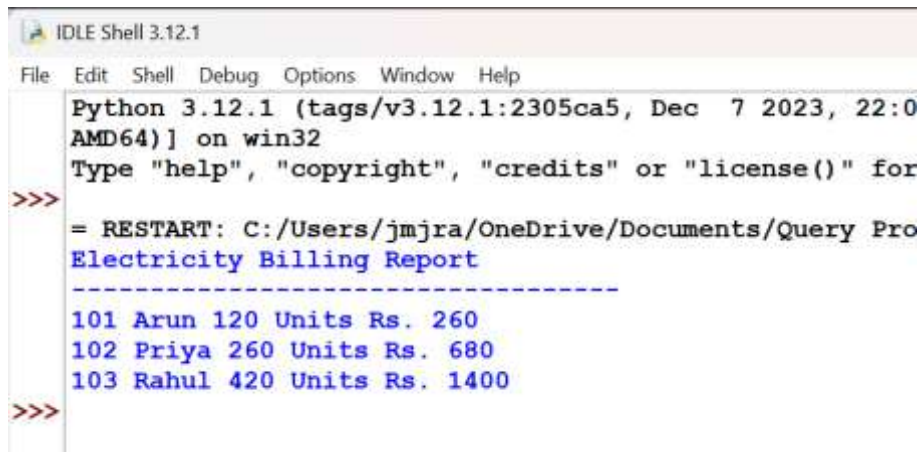
102,260

103,420

Code

```
import json
import csv
with open("consumers.json", "r") as f:
    consumers = json.load(f)
readings = {}
with open("readings.csv", "r") as f:
    reader = csv.DictReader(f)
    for row in reader:
        readings[int(row["id"])] = int(row["units"])
def bill(units):
    if units <= 100:
        return units * 2
    elif units <= 300:
        return 100 * 2 + (units - 100) * 3
    else:
        return 100 * 2 + 200 * 3 + (units - 300) * 5
print("Electricity Billing Report")
print("-" * 35)
for c in consumers:
    units = readings.get(c["id"], 0)
    amount = bill(units)
    print(c["id"], c["name"], units, "Units", "Rs.", amount)
```

Output



```
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec 7 2023, 22:0
AMD64) on win32
Type "help", "copyright", "credits" or "license()" for
>>>
= RESTART: C:/Users/jmjra/OneDrive/Documents/Query Pro
Electricity Billing Report
-----
101 Arun 120 Units Rs. 260
102 Priya 260 Units Rs. 680
103 Rahul 420 Units Rs. 1400
>>>
```

2. Hospital Emergency Services

A hospital stores ambulance information in XML format and emergency call records in CSV format. Write a Python program to assign ambulances to emergency requests based on availability and generate a dispatch report.

Aim

To assign available ambulances to emergency calls using ambulance data from an XML file and emergency requests from a CSV file.

Procedure

1. Read ambulance details from the XML file.
2. Read emergency call records from the CSV file.
3. Assign the first available ambulance to each emergency request.
4. Display the ambulance dispatch report.

Input

ambulances.xml

```
<ambulances>
  <ambulance>
    <id>A1</id>
    <driver>Ramesh</driver>
    <available>Yes</available>
```

```
</ambulance>

<ambulance>

  <id>A2</id>

  <driver>Suresh</driver>

  <available>Yes</available>

</ambulance>

<ambulance>

  <id>A3</id>

  <driver>Ganesh</driver>

  <available>No</available>

</ambulance>
```

```
</ambulances>
```

calls.csv

patient,location

John,T Nagar

Anita,Velachery

David,Anna Nagar

Code

```
import xml.etree.ElementTree as ET
import csv

tree = ET.parse("ambulances.xml")
root = tree.getroot()

available = []

for amb in root.findall("ambulance"):
    if amb.find("available").text == "Yes":
        available.append({
            "id": amb.find("id").text,
            "driver": amb.find("driver").text
        })
```

```

calls = []

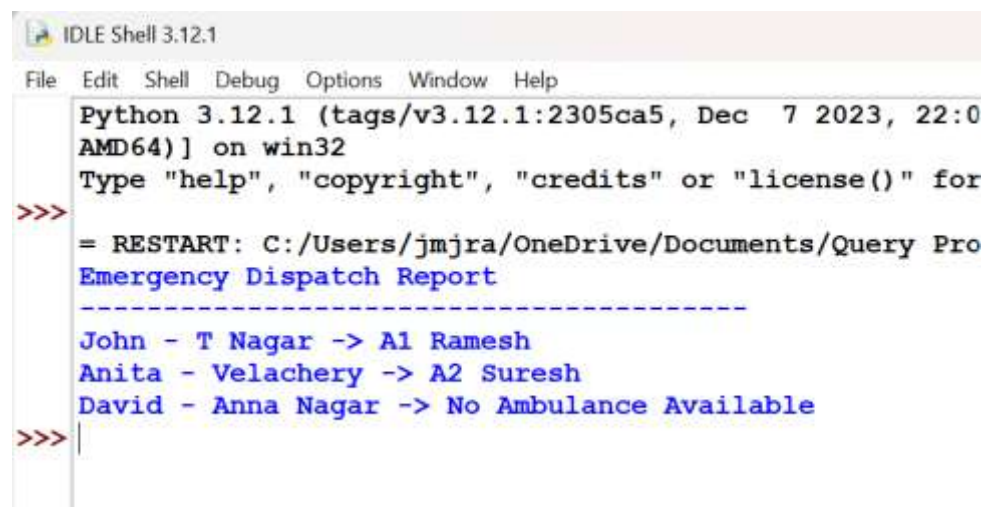
with open("calls.csv", "r") as f:
    reader = csv.DictReader(f)
    for row in reader:
        calls.append(row)

print("Emergency Dispatch Report")
print("-" * 40)

for call in calls:
    if available:
        amb = available.pop(0)
        print(call["patient"], "-", call["location"], "->", amb["id"], amb["driver"])
    else:
        print(call["patient"], "-", call["location"], "-> No Ambulance Available")

```

Output



```

IDLE Shell 3.12.1
File Edit Shell Debug Options Window Help
Python 3.12.1 (tags/v3.12.1:2305ca5, Dec 7 2023, 22:0
AMD64) on win32
Type "help", "copyright", "credits" or "license()" for
>>>
= RESTART: C:/Users/jmjra/OneDrive/Documents/Query Pro
Emergency Dispatch Report
-----
John - T Nagar -> A1 Ramesh
Anita - Velachery -> A2 Suresh
David - Anna Nagar -> No Ambulance Available
>>>

```